

CLC raster status data layers

100-m resolution raster CLC data were created by the rasterization of seamless European CLC vector data using CELL_CENTER method¹². Data are provided in compressed GeoTiff format (integer values, pixel depth: 8 bit).

Attributes provided for CLC raster status data

Attributes of the Raster Attribute Table are presented showing the example of CLC2018 status product in Table 3:

File name: U2018_CLC2018_V2020_20u1.tif

Table 3: Attributes for CLC raster status data

Attribute names	Explanation
OID	Built-in numeric unique ID of the records showing individual raster values
Value	Integer raster value showing indications of the CLC code in the range of 1,...44, corresponding to CLC codes 111,...523, plus code 48 corresponding to 999: NODATA (representing only the area of Andorra, which is not covered by CLC)
COUNT	Built-in numeric attributes showing the number of raster cells with the same raster value ¹³
LABEL3	Level-3 CLC class names as text
Red	Stored as real number in range 0..1; RGB red CLC colour code
Green	Stored as real number in range 0..1; RGB green CLC colour code
Blue	Stored as real number in range 0..1; RGB blue CLC colour code
CODE_18	Three-digit level-3 CLC code as text

CLC-Change products

CLC-Change layer products are available for the reference years of 2000 (partial EEA39 coverage), 2006, 2012 and 2018 as individual download packages. A new database version always include data covering the previous inventories. Data representing a previous inventory have the same structure, but geographic coverage might be different (see [AD05](#)). The content of a download package is explained by means of the example of CLC-Change₂₀₁₂₋₂₀₁₈ layer.

Base folder (example names listed for various formats)¹⁴:

- u2018_cha1218_v2020_20u1_fgdb (ESRI File Geodatabase vector format)
- u2018_cha1218_v2020_20u1_geopackage (SQLite Geopackage vector format)

¹² <https://desktop.arcgis.com/en/arcmap/latest/tools/conversion-toolbox/how-polygon-to-raster-works.htm>

¹³ In present case with 100m raster resolution the number is equivalent with the area expressed in hectares.

¹⁴ Directory names and file names below follow the file naming conventions described in Annex 2.

- u2018_cha1218_v2020_20u1_raster100m (100m resolution raster)

\DATA folder (contains raster or vector CLC-Change data layers):

- U2018_CHA1218_V2020_20u1.gdb (European seamless mosaic in ESRI File Geodatabase)
- U2018_CHA1218_V2020_20u1.gpkg (European seamless mosaic SQLite Geopackage)
- U2018_CHA1218_V2020_20u1.tif (European seamless mosaic in raster Geotiff format)

Additional folders have the same content as described under the CLC status layer products.

Description of CLC-Change vector data layers

Harmonization on country border was done only for CLC-Change₁₉₉₀₋₂₀₀₀ layer. For other layers only small sliver polygons along borders (<0,1 ha) have been dissolved. European CLC-Change seamless vector database represents the final vector product of European CLC-Change data integration, created in ESRI GeoDatabase format, and provided in SQLite Geopackage format as well.

Some redundant lines between neighbouring polygons with the same code are still present in database, but only as result of persisting ‘adaptive tiling’ procedure limitation of ArcGIS¹⁵. Polygons under MMU (5 ha) may be present along national borders and along tile boundaries. Polygons (slivers) under 0,1 ha may be present only along tile boundaries.

Feature attributes provided for CLC-Change vector data

The actual content of the Feature Attribute Table is slightly different for the two vector formats provided (ESRI Geodatabase/SQLite Geopackage). Attribute names are presented showing the example of CLC-Change₂₀₁₂₋₂₀₁₈ in Table 4:

File name: U2018_CHA1218_V2020_20u1.gdb / U2018_CHA1218_V2020_20u1.gpkg

Table 4: Attributes for CLC-Change vector data

Attribute names	Explanation
OBJECTID (or FID)	Built-in numeric unique ID of CLC polygon features
Shape	Built-in attribute showing the feature type (polygon)
Change	CLC change types as text (e.g. “211-112”)
ID	Unique ID of CLC change polygon features as text (e.g. “EU_1”)
CODE_12	Three-digit level-3 CLC code for the 2012 reference year as text
CODE_18	Three-digit level-3 CLC code for the 2018 reference year as text
Chtype	Text attribute with possible values “R” for real and “T” for technical changes
Remark	Remarks provided for individual CLC polygons (usually empty)
Area_Ha	Area of individual CLC polygons in hectares
Shape_length (ESRI Geodatabase only)	Built-in numeric attribute showing the perimeter of individual CLC polygons in meters

¹⁵ <https://desktop.arcgis.com/en/arcmap/latest/tools/supplement/tiled-processing-of-large-datasets.htm>

Shape_Area (ESRI Geodatabase only)	Built-in numeric attribute showing the area of individual CLC polygons in square meters
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Description of CLC-Change raster data layers

100-m resolution raster CLC-Change data were created by the rasterization of seamless CLC-Change vector data using CELL_CENTER method. In order to keep 8-bit resolution for change layers, they are divided into two files, representing consumption (from) and formation (to) part of change. Data are provided in compressed GeoTiff format (integer values, pixel depth: 8 bit).

Attributes provided for CLC-Change raster data

Attribute names are presented showing the example of CLC-Change₂₀₁₂₋₂₀₁₈ in Tables 5 and 6.

File name: U2018_CHA1218_12_V2020_20u1 - contains consumption part of change ('from' code).

Table 5: Attributes for CLC-Change raster data (consumption part)

Attribute names	Explanation
OID	Built-in numeric unique ID of the records showing individual raster values
Value	Integer raster value showing indications of the CLC code in the range of 1,...44, corresponding to CLC codes 111,...523, plus code 48 corresponding to 999: NODATA (representing only the area of Andorra, which is not covered by CLC)
COUNT	Built-in numeric attributes showing the number of raster cells with the same raster value
LABEL3	Three-digit level-3 CLC class names as text
CODE_12	Three-digit level-3 CLC code for the 2012 reference year as text
Red	Stored as real number in range 0..1; RGB red CLC colour code
Green	Stored as real number in range 0..1; RGB green CLC colour code
Blue	Stored as real number in range 0..1; RGB blue CLC colour code

File name: U2018_CHA1218_18_V2020_20u1 - contains formation part of change ('to' code).

Table 6: Attributes for CLC-Change raster data (formation part)

Attribute names	Explanation
OID	Built-in numeric unique ID of the records showing individual raster values
Value	Integer raster value showing indications of the CLC code in the range of 1,...44, corresponding to CLC codes 111,...523, plus code 48 corresponding to 999: NODATA (representing only the area of Andorra, which is not covered by CLC)
COUNT	Built-in numeric attributes showing the number of raster cells with the same raster value
LABEL3	Three-digit level-3 CLC class names as text
CODE_18	Three-digit level-3 CLC code for the 2012 reference year as text
Red	Stored as real number in range 0..1; RGB red CLC colour code

Green	Stored as real number in range 0..1; RGB green CLC colour code
Blue	Stored as real number in range 0..1; RGB blue CLC colour code

CLC accounting layers

Due to the technical characteristics of CLC and CLC-Change data, the evolution of the CLC update methodology and the quality of input data, time-series statistics derived directly from historical CLC data include several inconsistencies (see more in Ch. VIII. *How to use CLC?*). A harmonization methodology has been developed to create a statistically solid basis for CLC time series analysis by means of the so-called CLC accounting layers (more detail on account methodology is provided in Annex 5.) CLC accounting layers are “expert products” and are available following the link: <https://www.eea.europa.eu/data-and-maps/data/corine-land-cover-accounting-layers>

VI. Product methodology and workflow

Workflow of CORINE Land Cover inventories

CLC production is a good and well-functioning example of cooperation between European institutions (EEA and ESA), the Member States and industry. The workflow of the five, consecutive CLC inventories is rather similar, including nearly the same work packages. Table 7 provides an overview of the role of contributing partners involved in the execution of each work package in CLC2018.

Table 7: Work packages of a CORINE Land Cover inventory (case: CLC2018)

WP	Tasks	NRC / NT	EEA / ETC	ESA	Service Provider
1.1	Satellite image data acquisition (Sentinel-2 and Landsat-8)			X	
1.2	Ortho-correction			X	
1.3	Technical preparation of IMAGE2018	o			X
2	In-situ and ancillary data collection	X	X		
3.1	Mapping CLC Changes layer(2012-2018)	X	o		
3.2	Creation of CLC2018 status layer	X	o		
4	Verification by CLC Technical Team	o	X		
5	Final quality control and producing European products	o	X		
6	Validation				X
7	Data dissemination	X	X		

X = leading partner

o = partner involved

WP1 (satellite image provision) is dominantly centrally organised (ESA being responsible for image acquisition, Service Provider for technical preparation). WP2 (in-situ data) is shared by the EIONET National Reference Centres Land Cover (NRC) / National Teams (NT) and the EEA. WP3 (CLC production) is implemented by NT (National Teams in CLC2018 are listed in Annex 3) with support from EEA/ETC. WP4 (verification) is performed by the CLC Technical Team of EEA/ETC. WP5 (Final products) is implemented by the EEA/ETC. WP6 (validation) is performed by a Service Provider. WP7 (dissemination) is shared between the EEA and the NRC.

Satellite image acquisition and orthorectification

To fulfil the technical specifications of CLC, Earth observation data (satellite images) with the following characteristics are needed:

- resolution (pixel size): 10-30 metres
- spectral bands (optimal for landscape mapping): red, NIR, SWIR (supporting photointerpretation)
- coverage-1 (date of imagery optimal for CLC mapping): image acquisition window determined by the national team
- coverage-2 (date of imagery to provide optimal complimentary information, from CLC2006): 2nd image acquisition window determined by national team
- pre-processing level: orthorectified, national projection.

Satellite images to support CLC mapping are provided by ESA¹⁶. These image data sets are often referred to as IMAGE[year], where [year] refers to the CLC reference year (e.g. IMAGE2018 was mostly taken in 2017 with gap-filling in 2018, see Table 1). These satellite images are also usually used for purposes other than the CLC inventory (e.g. High-Resolution Layers). At the time of the first four CLC inventories, ESA did not have its dedicated satellite(s) suitable for continental land monitoring, therefore ESA established agreements (since 2009 in the frame of the GMES Space Component) with appropriate satellite image (typically Landsat, IRS, SPOT) providers for image acquisition. As the Sentinel-2 satellites have been placed on orbit in 2015 and 2017, they provided the primary data support for the fifth (CLC2018) and further inventories. In CLC2018, image acquisition (including thousands of images) and ortho-correction of satellite imagery (level-1C images in UTM/WGS84 projection) was provided by ESA through the Sentinel Hub.

It is usually a difficult task to cover the whole EEA39 area (6 Mkm²) with cloud-free imagery¹⁷. Therefore, for “gap-filling”, images of other satellite(s) having comparable characteristics are collected. In CLC2018 the image acquisition campaign was especially short, as the dedicated Sentinel-2 image acquisition campaign IMAGE2018 was confined to the year of 2017 (covering a single year, instead of 3 years of previous campaigns). Landsat-8 imagery was used as gap-filling to cover areas where no Sentinel-2 imagery was available in 2018 (Table 1).

Technical preparation of image products

In most of the CLC inventories, image enhancement techniques were applied in pre-processing of satellite imagery to support visual photointerpretation. The most common methodology applied is high-pass filtering in order to enhance geometric (structural) detail of the imagery, i.e. to improve interpretability (Chavez et al., 1991). The image product types available in CLC2018 ([AD03](#)) were as follows:

- The main – so-called “visual” - product, Sentinel-2 data (or Landsat-8 for gap filling) re-projected into national projection, was centrally produced by a Service Provider. This product

¹⁶ Some national land monitoring programmes are using different imagery, having higher-resolution. E.g. in Spain SPOT-5 (2,5 m pixel size), and in Germany RapidEye (5 m pixel size)

¹⁷ Cloud free means in practice no more than 5% cloud cover

included selected bands (red, NIR, SWIR) enhanced using the gradient image of the synthetic panchromatic band (in case of Sentinel-2) or the panchromatic band (Landsat-8).

- Additionally, the full, not-enhanced image products were provided by ESA, without modifications, for those countries that wanted to go beyond visual interpretation.

In-situ data

In-situ data (ancillary or reference data) in Copernicus programme, by definition, comprise all non-space-born data with a geographic dimension. Major use of in situ data in CLC project is to complement the satellite data in the course of production and to verify or validate results provided from space-born data. In-situ data is often used for provision of land use, crop type and habitat information, among others. Most frequently used and most needed in-situ data include:

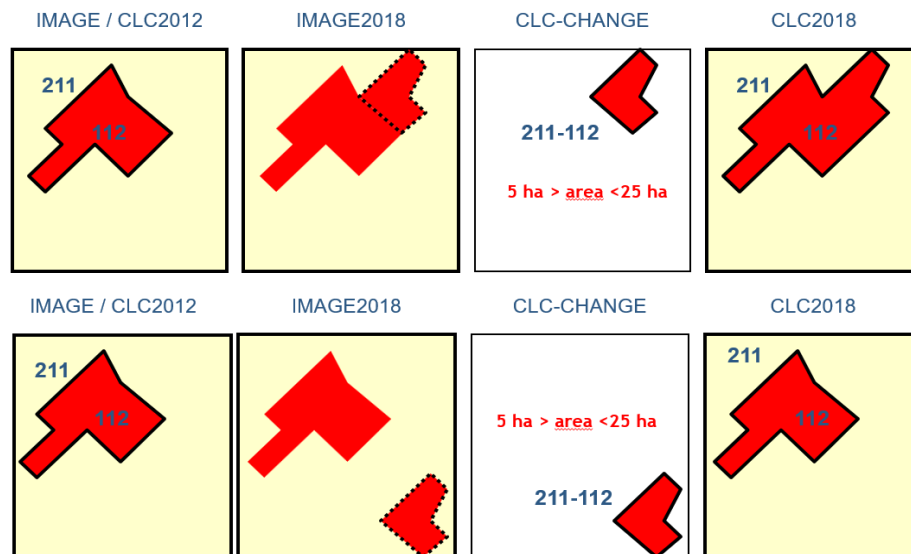
- Up-to date topographic maps (preferably at scale 1:25.000 / 1:50.000) to be used during interpretation, mapping and validation process.
- Orthophotos, taken optimally in the years when the satellite images were taken (2012 and 2017 in case of the CLC2018 project (especially if topographic maps are out of date).
- Thematic maps (built-up, vegetation type/habitats, forestry, hydrology, etc.).
- LPIS/IACS data, which have an utmost importance in precise mapping of agriculture classes and their changes.
- Other ancillary data (e.g. urban planning, nature conservation, forest fire) for identification/interpretation and verification of CLC classes and their changes.
- LUCAS field survey data coordinated by Eurostat, including landscape photographs from visited points, covering EU member states. The latest LUCAS survey was conducted in 2018 (Eurostat, 2019). LUCAS data are occasionally used in mapping, but more frequently in validation.
- High-Resolution Layers (HRL) of CLMS (especially degree of imperviousness and tree cover density / forest type). These data were available to support CLC2012 and CLC2018 (and produced by using IMAGE2012 and IMAGE2018 data).
- Ancillary data complementing the above and being useful in CLC change mapping are Google Earth (GE) imagery (or equivalent): it provides VHR image data, supporting the interpretation where no ortho-photos are available. Often multi-year time series are provided (time slider function), which are especially useful in understanding the evolution of an area. Crowd-sourced field photos, labels of different layers and StreetView function of GE can be useful, too in providing in-situ evidence or information on function of land/buildings or on natural vegetation.
- For the verification done by the CLC Technical Team often the only ancillary data are provided by GE.

Mapping CLC Changes

The term 'change layer' is used for CLC maps reflecting the change of land cover between two reference dates. Since the CLC2006 inventory, CLC-Change(year_{old}-year_{new}) is the primary product of the updating. It is produced directly (i.e. **not** derived by intersecting CLC[year_{old}] and CLC[year_{new}]) and

has a MMU of 5 ha, which means significantly more detail than the 25 ha the MMU of the status layers (Figure 4).

The aim of creating CLC-Change is to produce a map of real land cover changes describing an evolution process taking place in the environment (e.g. urban sprawl, forest clearcut). Changes should be interpreted regardless of their position (i.e. connected to an existing polygon or being “island”-like, see Figure 5)¹⁸.



Upper row: growth of an existing settlement. Lower row: birth of a new (isolated) settlement

- First boxes in both rows show the land cover status visible on IMAGE2012 and the polygon outlines in CLC2012 database.
- Second boxes show the land cover status visible on IMAGE2018 without polygon boundaries. Dashed outline marks patches that have changed.
- Third boxes show polygons to be drawn in the CLC-Change database.
- Fourth boxes show the polygons as present in CLC2018 database (as the results of GIS addition of CLC2012 and CLC-Change₂₀₁₂₋₂₀₁₈).

Figure 5: Consistent mapping of CLC Change (AD03)

Change polygons should

- have size at least 5 ha,
- have width at least 100 m,
- describe a real evolution process that occurred between year_{old} and year_{new}, and
- be detectable on satellite images (with support of dedicated in-situ data, if needed).

There are two basic approaches of creating a change layer, ‘CLC updated first’ and ‘change mapping first’, summarized on Figure 6.

¹⁸ In CLC2000 island-like changes having size between 5 and 25 ha were not interpreted.

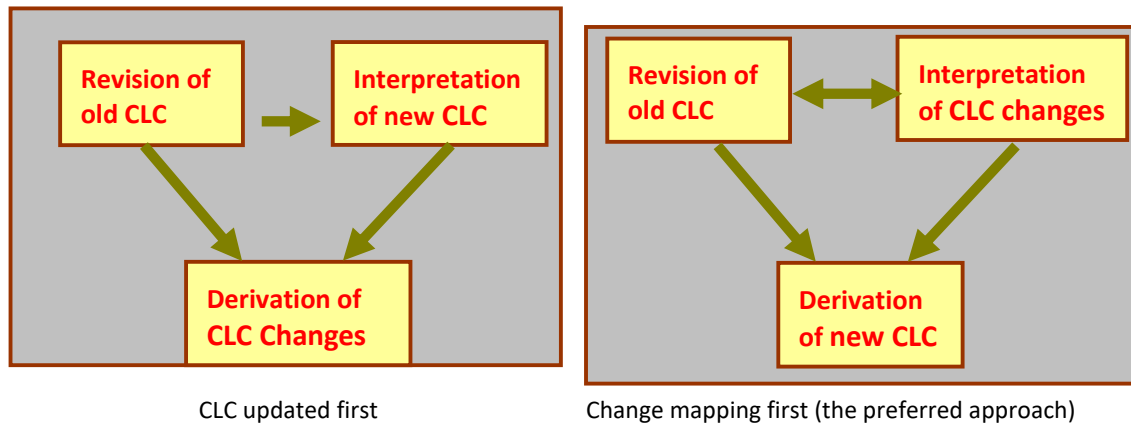


Figure 6: Approaches for CLC update (AD03)

The 'change mapping first' methodology supports better the above goals than the 'update first' approach (Figure 6) as (1) changes are interpreted directly (the interpreter has to think about what the real process was); and (2) all changes larger than 5 ha can be easily delineated regardless of their geometric position (whether attached to an existing polygon or not). Isolated changes (island-like polygons) having size between 5 and 25 ha cannot be mapped using the 'update first' method (Büttner, 2014).

Understanding CLC Changes

CLC changes can include those changes of the Earth surface that have longer than yearly/seasonal periodicity. Urban sprawl, plantation of olive trees to replace arable land, melting of glaciers or creation of a new water reservoir are such long-term land cover changes to be mapped as CLC change. On the other hand, transient changes and short-term periodical changes are phenomena not to be mapped as CLC change.

Transient changes (lasting only for short time), not to be mapped as CLC change are e.g.:

- Changes along rivers due to floods (temporary inundations).
- Changes inland due to heavy rains (temporary water-logging).

What to do with periodical changes, where LC status alternates between two different land covers? Several such processes have periodicity shorter than 1 year, therefore are not to be considered as CLC change:

- Water level changes in coastal areas due to tidal phenomena (high tide, low tide).
- Changes in lake / wetland area due to seasonal water level differences.
- Intermittent water cover of glacial rivers during the spring snow-melt period.
- Regular changes of water cover of fishponds due to maintenance.
- Seasonal phenological changes in forests (status of leaves).
- Seasonal phenological changes of natural grassland and sparse vegetation (green in spring, yellow in summer).